How I Choose the Right Chart

Data Analyst study notes | Beginner level | Simple steps, simple words

What is a chart, really? A chart is just an **answer to a question**, drawn as a picture. If I do not know the question, I cannot pick the chart. So I always start with the question, then look at my data, then choose.

My 4-step method

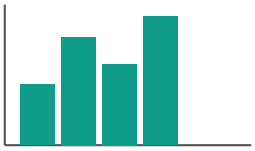
1	**Ask the question**	What do I want my reader to know? Example: *"Which product category earns the most?"* Write it in one line first.
2	**Look at my data**	Is it a **category** (product, city), a **number** (sales, price) or a **date** (month, year)? The type of data limits which charts will work.
3	**Match question to chart**	Use the table below. Comparing? Bar. Trend over time? Line. Share of total? Pie or stacked bar. Two numbers together? Scatter.
4	**Clean up and check**	Add a clear title, label the axes, use 1 to 2 colours, remove clutter. Then ask: *"Can a new person understand this in 5 seconds?"*

Question to chart: the main table

I want to...	Best chart	Simple example	Tip from my notes
Compare categories	**Bar / column chart**	Sales by product category	*Sort bars big to small*
See a **trend** over time	**Line chart**	Monthly revenue for 2 years	*Time always goes left to right*
Show **part of a whole**	**Pie / donut (max 4 slices) or stacked bar**	Share of sales by region	*More than 5 slices? Use a bar*
Find a **relationship**	**Scatter plot**	Price vs quantity sold	*Each dot = one item or customer*
See the **spread** of values	**Histogram or box plot**	Order value: most are small, a few are huge	*Great for spotting outliers*
Show a **ranking**	**Sorted horizontal bar**	Top 10 customers	*Put the biggest on top*
Show **one key number**	**KPI card**	Total revenue, total orders	*Add a small change vs last month*
Show **steps and drop-off**	**Funnel chart**	Visit, cart, checkout, order	*Each step is smaller than the last*
Show **places**	**Map**	Sales by state	*Only if location really matters*
Look up **exact values**	**Table**	Invoice list, detailed report	*Use when numbers matter more than shape*

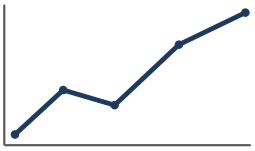
Mentor note: *Beginners often start with "which chart looks cool?" Seniors start with "what decision will this help someone make?" A plain bar chart that answers the question beats a fancy chart that confuses people.*

The 6 charts I use most (quick picture guide)



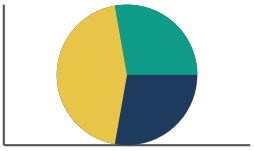
Bar / Column

Compare categories. Easiest chart to read. Start the axis at zero.



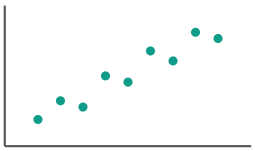
Line

Show change over time. Use for days, months, years.



Pie / Donut

Share of a total. Keep to 3 or 4 slices that add up to 100%.



Scatter

See if two numbers move together (up with up = positive link).

Histogram

See how values are spread. Shows where most data sits.

KPI Card

One big number. Perfect at the top of a dashboard.

Common mistakes (and what I do instead)

Mistake	Why it hurts	Do this instead
Pie chart with 8+ slices	Eyes cannot compare small slices	Use a sorted bar chart
3D charts	3D bends the shapes, so values look wrong	Stay flat and simple
Bar axis not starting at zero	Small gaps look huge, which misleads people	Start bars at 0 (lines may be flexible)
Too many colours	Reader does not know where to look	1 main colour + 1 highlight colour
Line chart for categories	A line suggests order or time between points	Use bars for categories
No title or axis labels	Reader must guess what they are seeing	Title = the message, e.g. "Electronics earns 40% of sales"

Quick decision cheat sheet

Is it about **time**?	Yes: **Line chart**
Comparing **groups**?	Yes: **Bar chart**
Showing **share of total**?	Yes: **Donut or stacked bar** (few parts only)
Two numbers, is there a **link**?	Yes: **Scatter plot**
Want to see the **spread**?	Yes: **Histogram**
Just **one number**?	Yes: **KPI card**
Still not sure?	Try a **bar chart**. It is almost never wrong.

5-second test before I share any chart: *(1) Does the title say the main message? (2) Can I read the axes? (3) Is there anything I can delete? (4) Would my manager understand it without me explaining? If all four are yes, the chart is ready.*